MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

GENERATOR, FUNCTION, LSB, ANALOG, KEY CODE SHU

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic LSB function generator module.

* 1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 202610-20T16:32Z
Check the source to verify that this is the current version before use.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.58 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be SHU.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/187-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc and ±15.0 ±1.5 V dc, when driving the specified loads.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

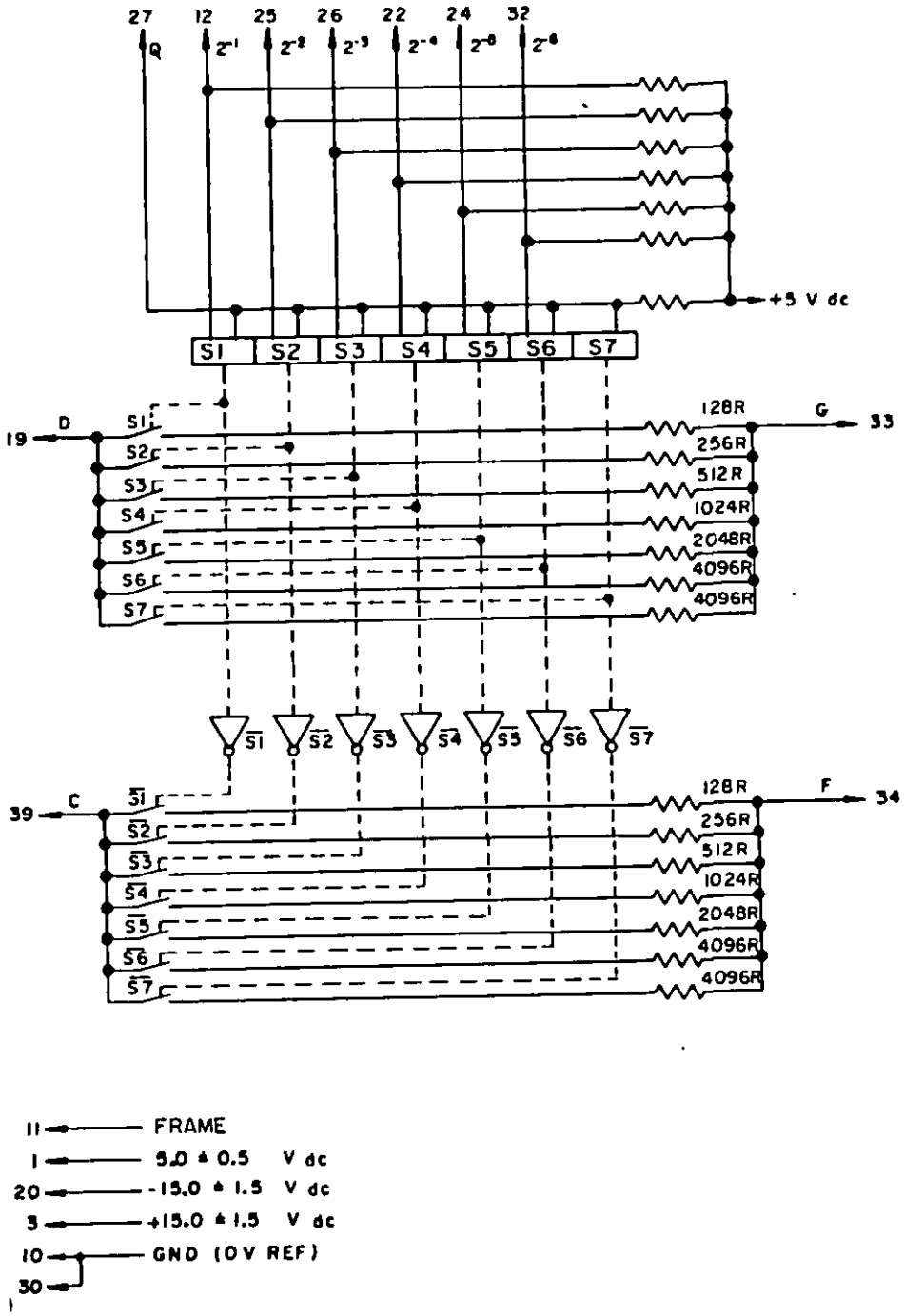
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

Source: https://assist.dla.mil -- Downloaded: 202640-20T16:32Z
Check the source to verify that this is the current version before use.



NOTE: The following pins are unused: 2, 4, 5, 6, 7, 8, 9, 13, 14, 15, 16, 17, 18, 21, 23, 28, 29, 31, 35, 36, 37, 38, and 40.

FIGURE 1. SHU module function diagram, LSB function generator.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:32Z
Check the source to verify that this is the current version before use.

*TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A24	Isolation	II	S	10	∞	1	∞	10	∞	1	∞	MΩ
B1-B7	Input breakdown current	III	S	0	1	0	1	0	1	0	1	mA
C1-C6	Hot socket check	IV	Q	-0.70	0.70	-0.70	0.70	-0.70	0.70	-0.70	0.70	Degrees of selected angle
D1-D14	Switch check	V	Q	◄		See figure 10, note 2						►
E1,E2	V_S stress check	VI	Q	-0.70	0.70	-0.70	0.70	-0.70	0.70	-0.70	0.70	Degrees of selected angle
F1-F24	Angular accuracy (400 Hz) check	VII	X	-0.70	0.70	-0.70	0.70	-0.70	0.70	-0.70	0.70	Degrees of selected angle
G1-G18	DC integrity and leakage check	VIII	X	◄		See footnote 4 and table VIII						►
H1	Absolute transformation ratio check	IX	X	44.8564	45.1429	44.8564	45.1429	44.1500	45.8500	44.1500	45.8500	Degrees
H2	Phase shift	IX	X	179.5	180.5	179.5	180.5	179.5	180.5	179.5	180.5	Degrees
J1-J4	Input current (analog)	X	X	0	50	0	50	0	50	0	50	µA
K1-K6	Input current, low level	XI	X	0.9	1.4	0.9	1.4	0.9	1.4	0.9	1.4	mA
K7	Input current, low level	XI	X	225	275	225	275	225	275	225	275	µA
K8-K13	Input current, high level	XI	X	275	440	275	440	275	440	275	440	µA
K14	Input current, high level	XI	X	68	84	68	84	68	84	68	84	µA
L1	Power supply current, 5.5 V	XII	S	0	3	0	3	0	3	0	3	mA
L2,L3	Power supply current, ±16.5 V	XII	S	0	5	0	5	0	5	0	5	mA
L4	Power supply current, 5.5 V	XII	Q	6.5	15.0	6.5	15.0	6.5	15.0	6.5	15.0	mA
M1	Continuity, ground	XIII	Q	0.0	0.1	0.0	0.1	0.0	0.1	0.0	0.1	Ω
M2	Continuity, ground	XIII	X	0	1	0	1	0	1	0	1	Ω

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T16:32Z
Check the source to verify that this is the current version before use.

*TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
M3	Continuity, frame	XIII	X	0	1	0	1	0	1	0	1	Ω
N1-N4	Settling time	XIV	X	0	4	0	4	0	4	0	4	µs
P1-P3	Angular accuracy (800 Hz) check	XV	Q	-0.70	0.70	-0.70	0.70	-0.70	0.70	-0.70	0.70	Degrees of selected angle
Q1-Q7	Input capacitance	XVI	Q	2	35	2	35	2	35	2	35	pF
Q8,Q9	Output capacitance	XVI	Q	50	200	50	200	50	200	50	200	pF
R1-R6	Power supply noise rejection	XVII	Q	0	1	0	1	0	1	0	1	mV rms
R7-R12	Power supply noise rejection	XVII	Q	0	6	0	6	0	6	0	6	mV rms

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ The initial and end-of-life limits for 25°C and operating temperature are the same.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:32Z
Check the source to verify that this is the current version before use.

4.4.2 Order of testing. Test conditions A through R shall be performed in any order with the following exceptions. If test conditions A, B, C, D, or E are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, E where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through R in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions F through R shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 6. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3. Following the life test and while still at the maximum class temperature, electrical tests F through R shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests F through R shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Cross reference of classification. Part number M28787/187-1 supersedes part number M28787/187. The two parts are interchangeable.

* 6.4 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life limits from table I.

Maximum power dissipation	
+5.5 V (15 mA)	82.5 mW
+16.5 V (5 mA)	82.5 mW
-16.5 V (-5 mA)	82.5 mW
Supply voltages	5 V dc ±10 percent and ±15 V dc ±10 percent

Source: https://assist.dla.mil - Downloaded: 2026-10-20T16:32Z
Check the source to verify that this is the current version before use.

Digital inputs (internally pulled-up to V_{CC})	
2^{-n} input current	
Low level (V_{IL} = 0 V)	-0.9 mA minimum, -1.4 mA maximum
High level (V_{IH} = 3.5 V)	-275 µA minimum, -440 µA maximum
High level (V_{IH} = 5.5 V)	1 mA maximum
Q input current	
Low level (V_{IL} = 0 V)	-225 µA minimum, -275 µA maximum
High level (V_{IH} = 3.5 V)	-68 µA minimum, -84 µA maximum
High level (V_{IH} = 5.5 V)	1 mA maximum
2^{-n} and Q input voltage	
Low level (V_{IL})	0.0 V minimum, 0.8 V maximum
High level (V_{IH})	3.5 V minimum, V_{CC} maximum
Settling time	4 µs maximum
Analog input voltage range	0.6 V minimum, 9.6 V maximum
S/D and D/S applications	
Analog input voltage	+0.6 V to +9.6 V
Analog reference voltage (typical)	4.6 V dc
Analog input current (V_I = 4.6 V)	-50 µA maximum
Switch dc offset voltage	1.5 mV maximum
Phase shift	±0.5 degrees
Angular accuracy (400 and 800 Hz)	±0.70 degrees (±0.33 arc minutes referenced to converter output)

* 6.5 Module function. This module contains multipliers consisting of semiconductor switches controlled by binary digital inputs that switch resistors in or out from a precision resistor network. This module provides the necessary digital-to-analog function used in digital-to-synchro and synchro-to-digital converter applications.

6.5.1 Input to output relationship. Each LSB function generator has the input to output relationship of figure 2.

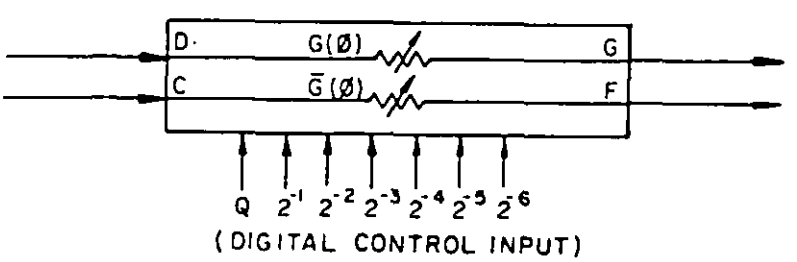


FIGURE 2. Input to output relationship block diagram.

Source: https://assist.dla.mil -- Downloaded: 202640-20T16:32Z
Check the source to verify that this is the current version before use.

The functions Sin θ and Cos θ are generated by the use of a nonlinear set of resistive ladder networks interconnected by the closure of solid state switching elements which in turn are actuated by the digital control inputs.

The proportionality inherent in the module is

$$G(\emptyset)_{D-C} = 0.001 \left[\overline{Q} \left(\sum_{i=1}^{6} \frac{\overline{a}_i}{2^{6+i}} + \frac{1}{2^{12}} \right) + Q \sum_{i=1}^{6} \frac{a_i}{2^{6+i}} \right] \text{ siemens (mhos)}$$

$$G(\emptyset)_{C-F} = 0.001 \left[Q \left(\sum_{i=1}^{6} \frac{\overline{a}_i}{2^{6+i}} + \frac{1}{2^{12}} \right) + \overline{Q} \sum_{i=1}^{6} \frac{a_i}{2^{6+i}} \right] \text{ siemens (mhos)}$$

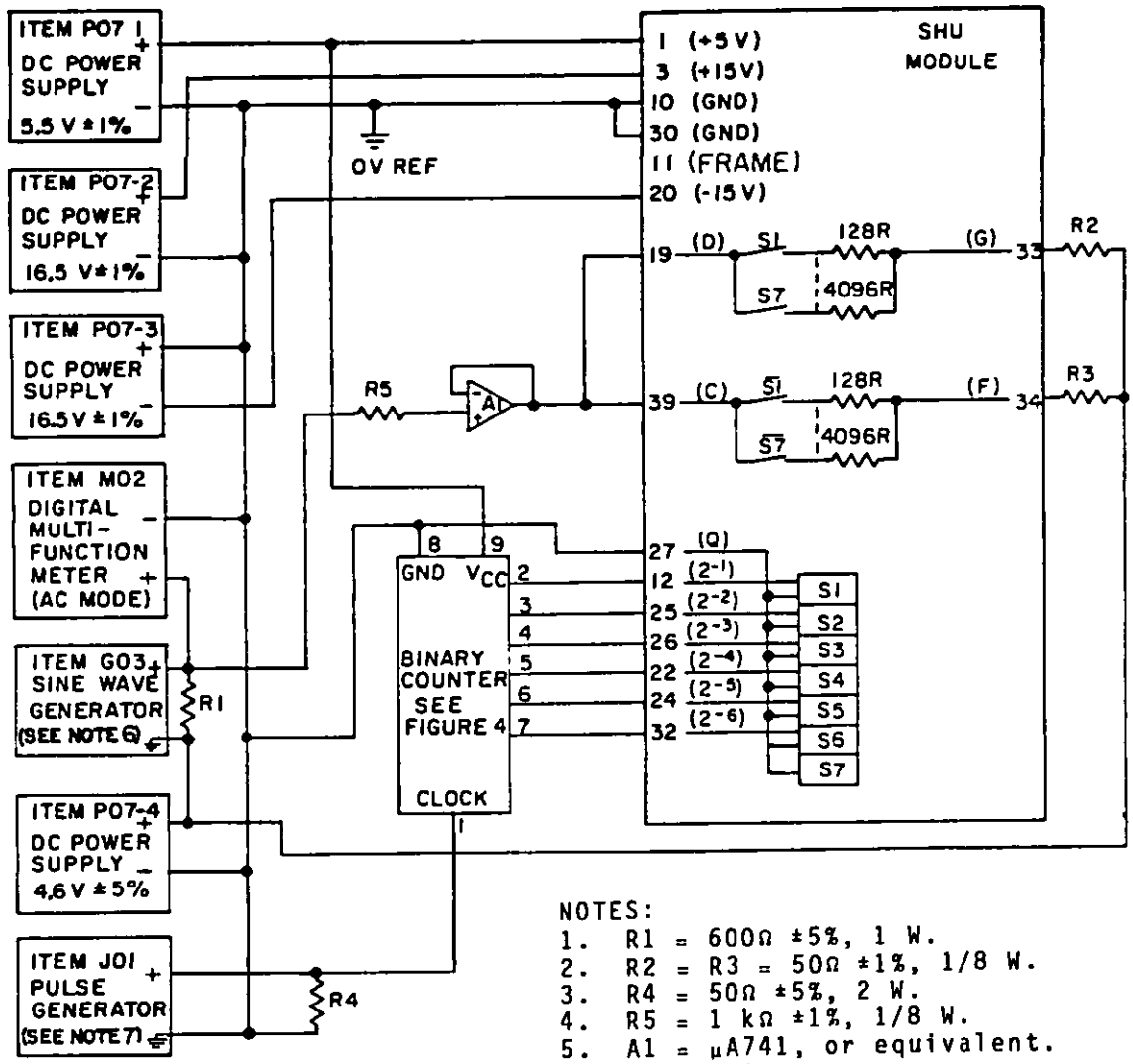
6.6 *Changes from previous issue.* The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5963-N543)

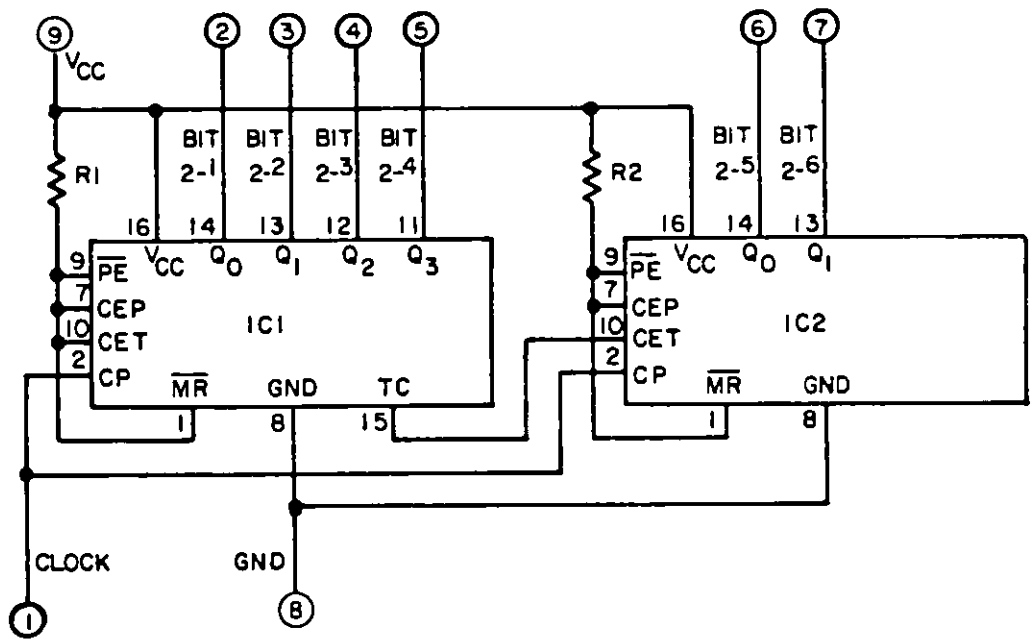
Source: https://assist.dla.mil -- Downloaded: 202640-20T16:32Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 600Ω ±5%, 1 W.
2. R2 = R3 = 50Ω ±1%, 1/8 W.
3. R4 = 50Ω ±5%, 2 W.
4. R5 = 1 kΩ ±1%, 1/8 W.
5. A1 = μA741, or equivalent.
6. Item G03 setup:
   Amplitude = 3.18 V rms ±1%.
   Frequency = 400 ±4 Hz.
7. Item J01 setup:
   Amplitude = 4 V ±10%.
   PW = 200 ±20 ns.
   PRR = 1.0 ±0.1 MHz.
   $t_{TLH} = t_{THL} = 25 \pm 20$ ns.

*FIGURE 3. Operating temperature and life test circuit.

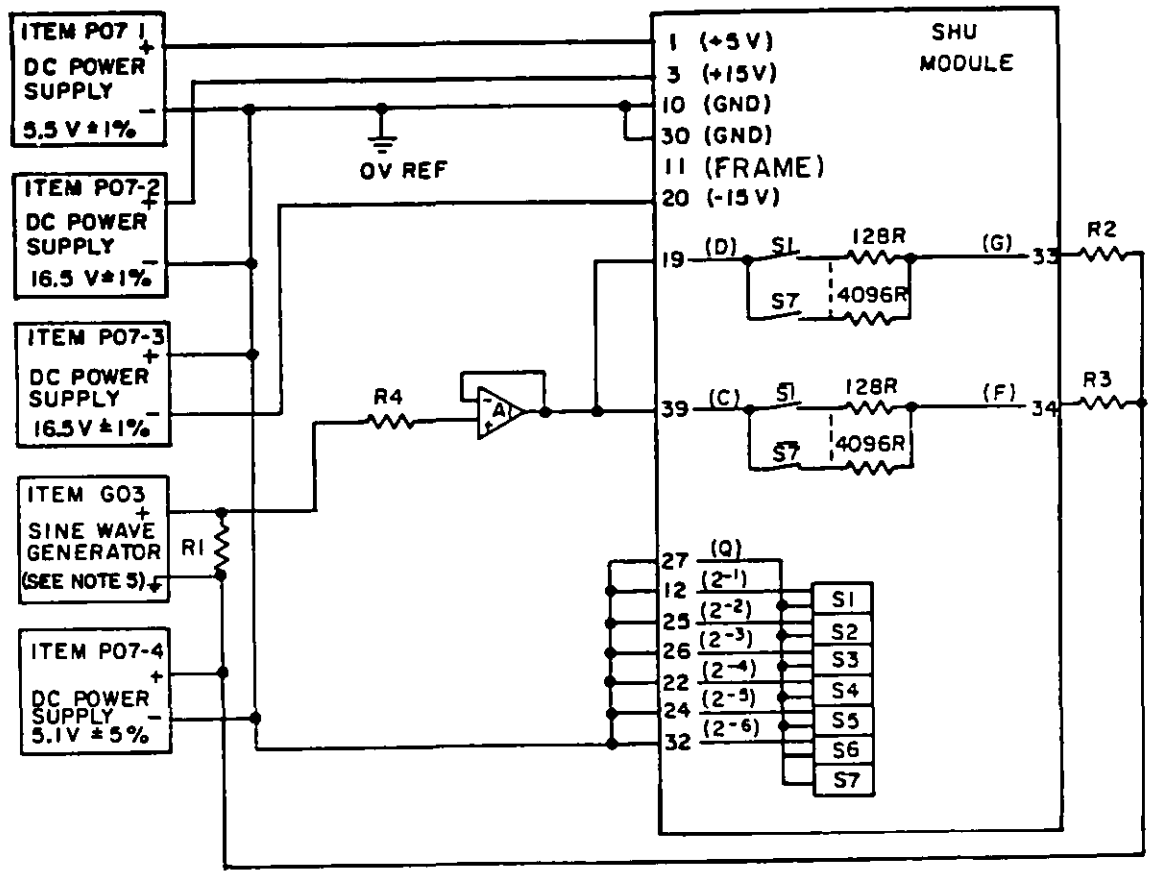
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:32Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = R2 = 2.2 kΩ ±5%, 1/8 W.
2. IC1 = IC2 = 9316, or equivalent.
3. V_{CC} = 5.0 ±0.5 V.

FIGURE 4. Binary counter.

Source: https://assist.dla.mil -- Downloaded: 202640-20T16:32Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 600Ω ±5%, 1 W.
2. R2 = R3 = 50Ω ±1%, 1 W.
3. R4 = 1 kΩ ±5%, 1/8 W.
4. A1 = μA741, or equivalent.
5. Item G03 setup:
   Amplitude = 3.18 V rms ±1%.
   Frequency = 400 ±4 Hz.
6. Repeat test with pin 27 open.

FIGURE 5. Component temperature test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:32Z
Check the source to verify that this is the current version before use.

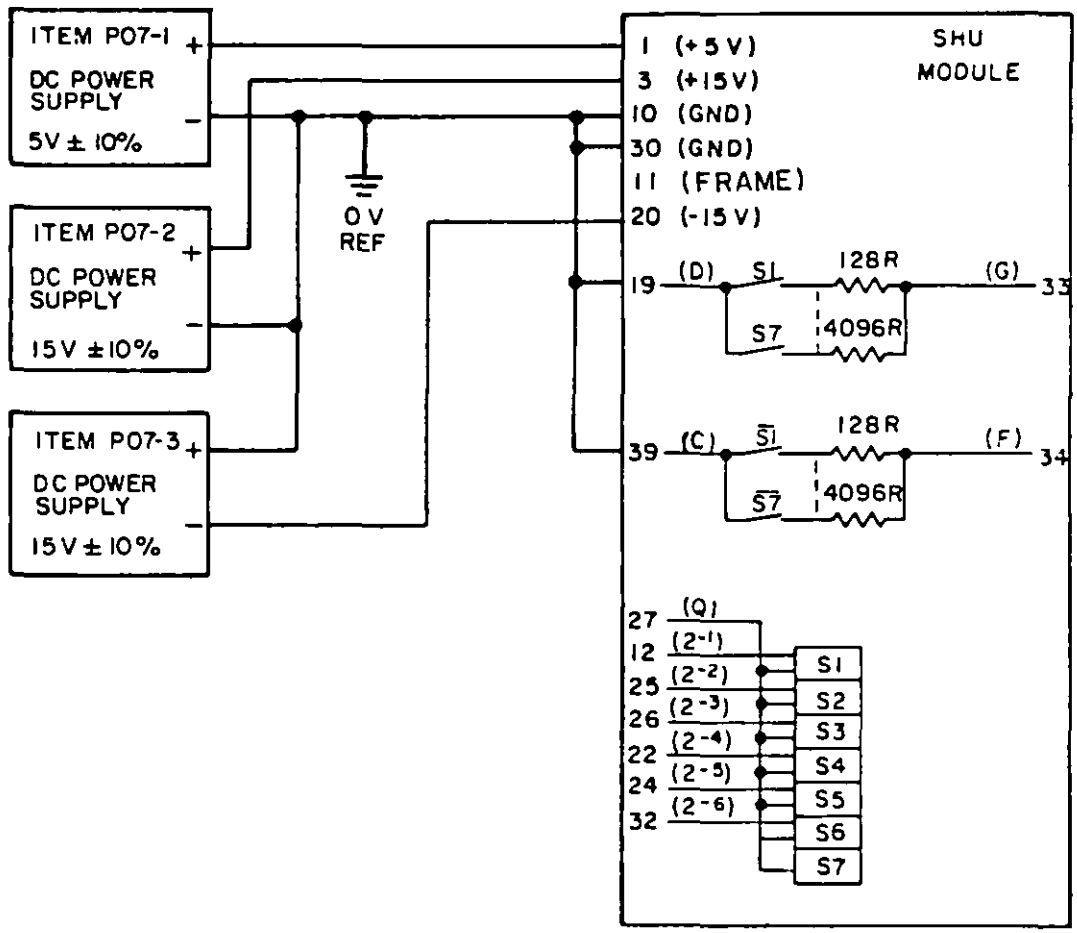


FIGURE 6. Vibration test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:32Z
Check the source to verify that this is the current version before use.

TABLE II. Isolation test.

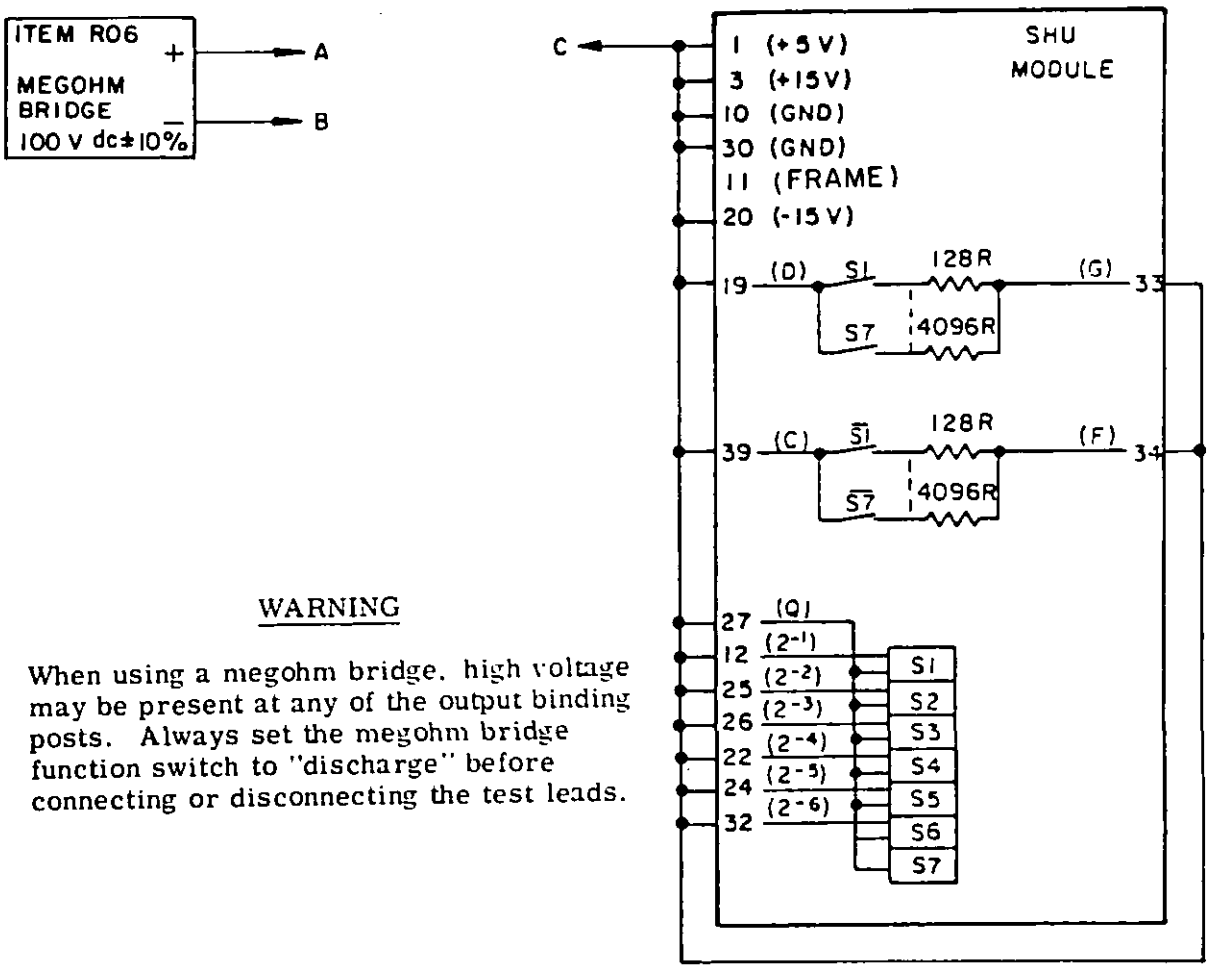
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation, module pins to frame	C,2,4,5,6,7,8,9,13,14, 15,16,17,18,21,23,28,29, 31,35,36,37,38,40	11	10	∞	MΩ	7
A2	Isolation, unused to used pins	2,4,5,6,7,8,9,13,14,15, 16,17,18,21,23,28,29, 31,35,36,37,38,40	C	"	"	"	"
A3	Isolation, unused to unused pins	2,4,5,6,7,8,9,13,14,15, 16,17,18,21,23,28,29, 31,35,36,37,38	40	"	"	"	"
A4	"	2,4,5,6,7,8,9,13,14,15, 16,17,18,21,23,28,29, 31,35,36,37	38	"	"	"	"
A5	"	2,4,5,6,7,8,9,13,14,15, 16,17,18,21,23,28,29, 31,35,36	37	"	"	"	"
A6	"	2,4,5,6,7,8,9,13,14,15, 16,17,18,21,23,28,29, 31,35	36	"	"	"	"
A7	"	2,4,5,6,7,8,9,13,14,15, 16,17,18,21,23,28,29, 31	35	"	"	"	"
A8	"	2,4,5,6,7,8,9,13,14,15, 16,17,18,21,23,28,29,	31	"	"	"	"
A9	"	2,4,5,6,7,8,9,13,14,15, 16,17,18,21,23,28	29	"	"	"	"
A10	"	2,4,5,6,7,8,9,13,14,15, 16,17,18,21,23	28	"	"	"	"
A11	"	2,4,5,6,7,8,9,13,14,15, 16,17,18,21	23	"	"	"	"
A12	"	2,4,5,6,7,8,9,13,14,15, 16,17,18	21	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:32Z
Check the source to verify that this is the current version before use.

TABLE II. Isolation test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A13	Isolation, unused to unused pins	2,4,5,6,7,8,9,13,14, 15,16,17	18	10	∞	MΩ	7
A14	"	2,4,5,6,7,8,9,13,14, 15,16	17	"	"	"	"
A15	"	2,4,5,6,7,8,9,13,14, 15	16	"	"	"	"
A16	"	2,4,5,6,7,8,9,13,14	15	"	"	"	"
A17	"	2,4,5,6,7,8,9,13	14	"	"	"	"
A18	"	2,4,5,6,7,8,9	13	"	"	"	"
A19	"	2,4,5,6,7,8	9	"	"	"	"
A20	"	2,4,5,6,7	8	"	"	"	"
A21	"	2,4,5,6	7	"	"	"	"
A22	"	2,4,5	6	"	"	"	"
A23	"	2,4	5	"	"	"	"
A24	"	2	4	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:32Z
Check the source to verify that this is the current version before use.



WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

FIGURE 7. Isolation test circuit for table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:32Z
Check the source to verify that this is the current version before use.

TABLE III. Input breakdown current test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	Q Input breakdown current	27	12,22,24,25,26,32	0	1	mA	8
B2	2^{-1} "	12	22,24,25,26,27,32	"	"	"	"
B3	2^{-2} "	25	12,22,24,26,27,32	"	"	"	"
B4	2^{-3} "	26	12,22,24,25,27,32	"	"	"	"
B5	2^{-4} "	22	12,24,25,26,27,32	"	"	"	"
B6	2^{-5} "	24	12,22,25,26,27,32	"	"	"	"
B7	2^{-6} "	32	12,22,24,25,26,27	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:32Z
Check the source to verify that this is the current version before use.

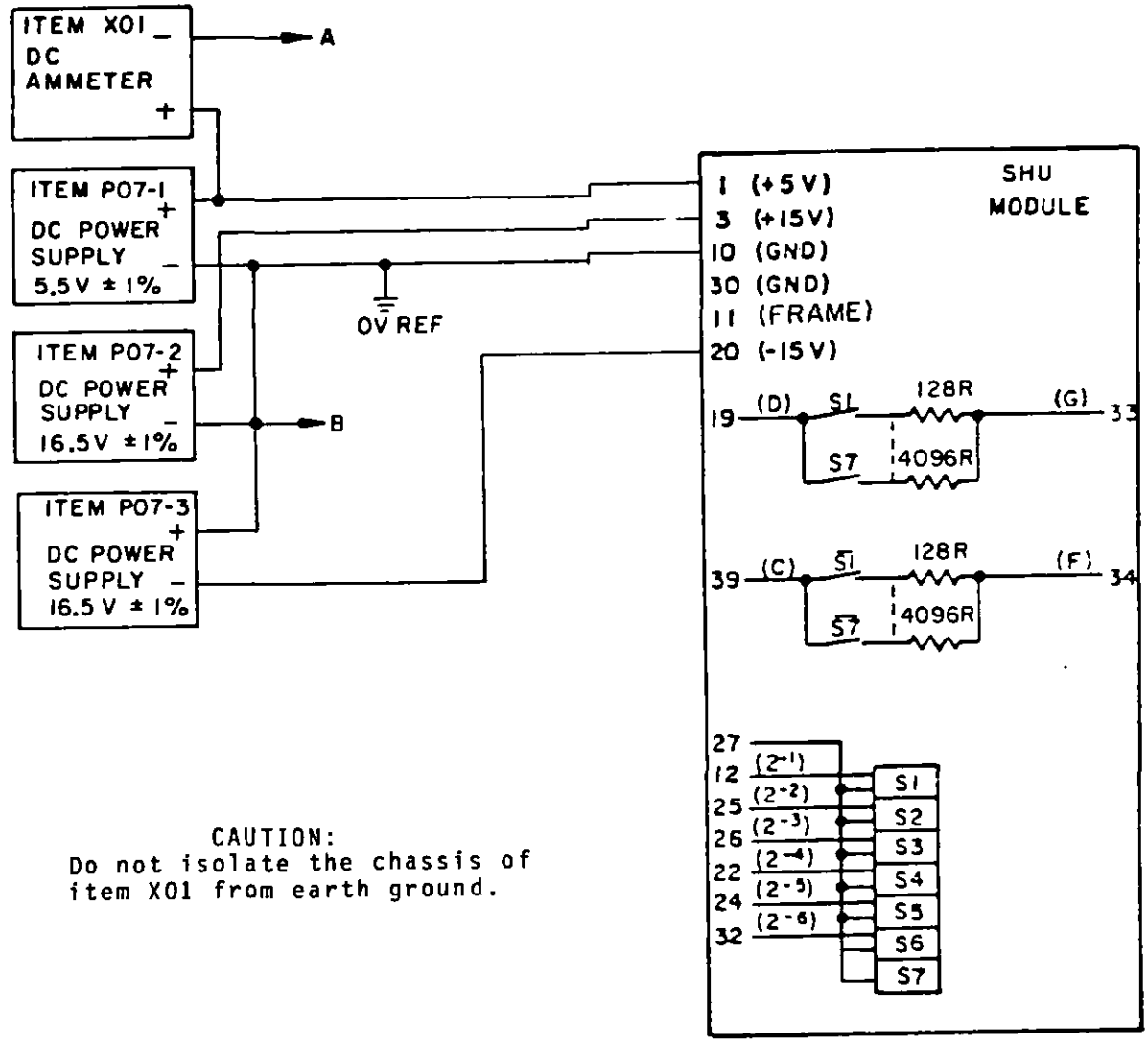


FIGURE 8. Input breakdown current test circuit for table III.

Source: https://assist.dla.mil Downloaded: 2026-10-20T16:32Z
Check the source to verify that this is the current version before use.

TABLE IV. Hot socket check test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	Hot socket check	27	Open	26.95	28.35	Degrees	9
C2	"	Open	27	61.65	63.05	"	"
C3	"	27	Open	26.95	28.35	"	"
C4	"	Open	27	61.65	63.05	"	"
C5	"	27	Open	26.95	28.35	"	"
C6	"	Open	27	61.65	63.05	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:32Z
Check the source to verify that this is the current version before use.

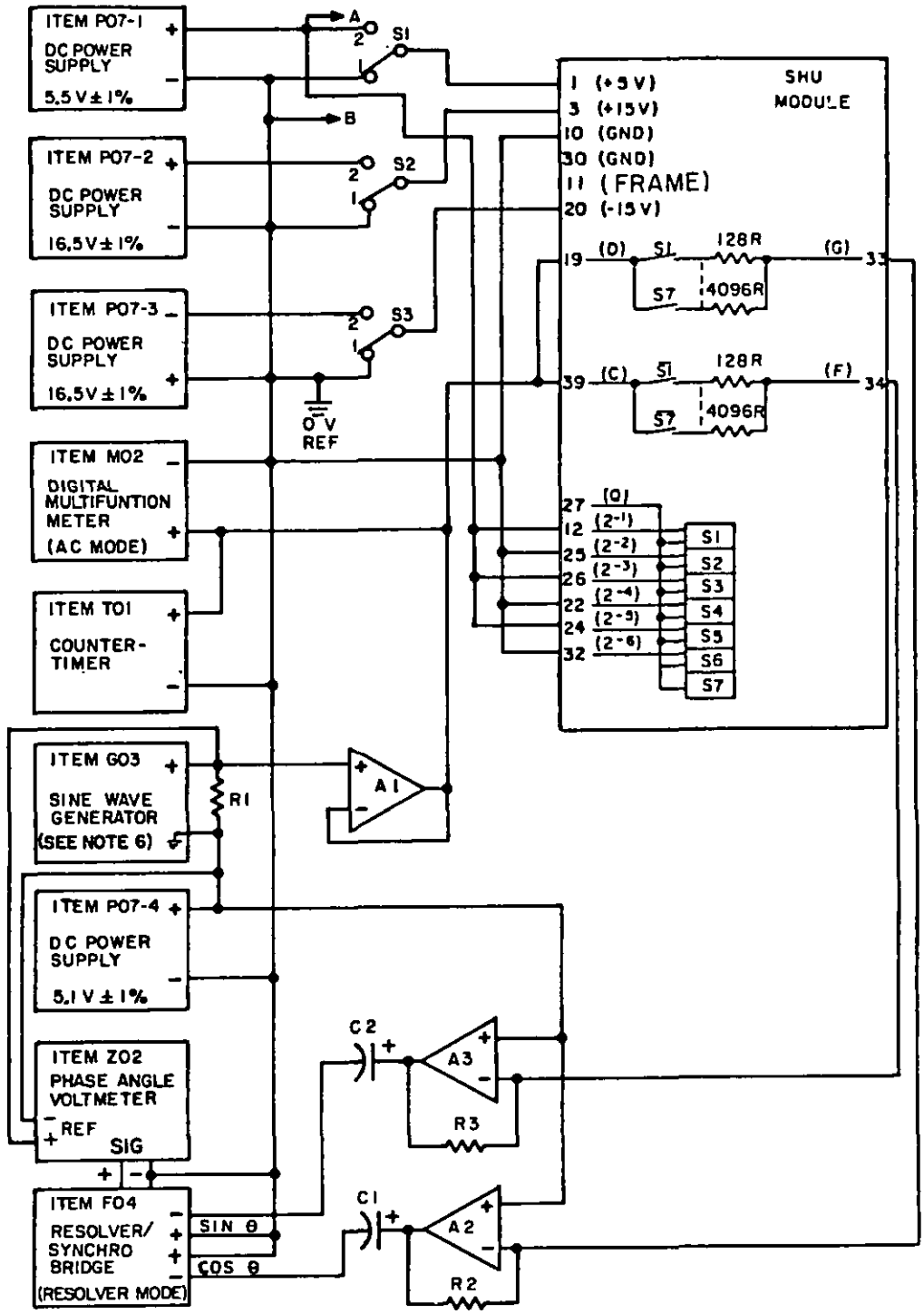


FIGURE 9. Hot socket check test circuit for table IV.

Source: https://assist.dla.mil - Downloaded: 202610-20T16:32Z
Check the source to verify that this is the current version before use.

NOTES:

1. R1 = 600Ω ±5%, 1 W.
2. R2 = R3 = 64 kΩ ±0.02% (matched to ±0.005%), 1/8 W.
3. C1 = C2 = 100 µF ±5%, 20 WV dc, electrolytic.
4. A1-A3 = µA741, or equivalent.
5. S1-S3 = SPDT switch.
6. Item G03 setup (verified at pins 19 and 39):
   Amplitude = 3.2 ±0.1 V rms
   Frequency = 400 Hz ±1%.
7. Switching sequence:
   (a) For test condition C1, set switches S1, S2, and S3 to position 1, then operate switch S1 to position 2, S2 to position 2, and S3 to position 2 *in this specified sequence.*
   (b) Test condition C2 must be performed following test condition C1 without power disruption.
   (c) For test condition C3, set switches S1, S2, and S3 to position 1, then operate switch S2 to position 2, S3 to position 2, and S1 to position 2 in this specified sequence.
   (d) Test condition C4 must be performed following test condition C3 without power disruption.
   (e) For test condition C5, set switches S1, S2, and S3 to position 1, then operate switch S3 to position 2, S1 to position 2, and S2 to position 2 in this specified sequence.
   (f) Test condition C6 must be performed following test condition C5 without power disruption.
8. This test verifies an angular accuracy of ±0.70 degrees.

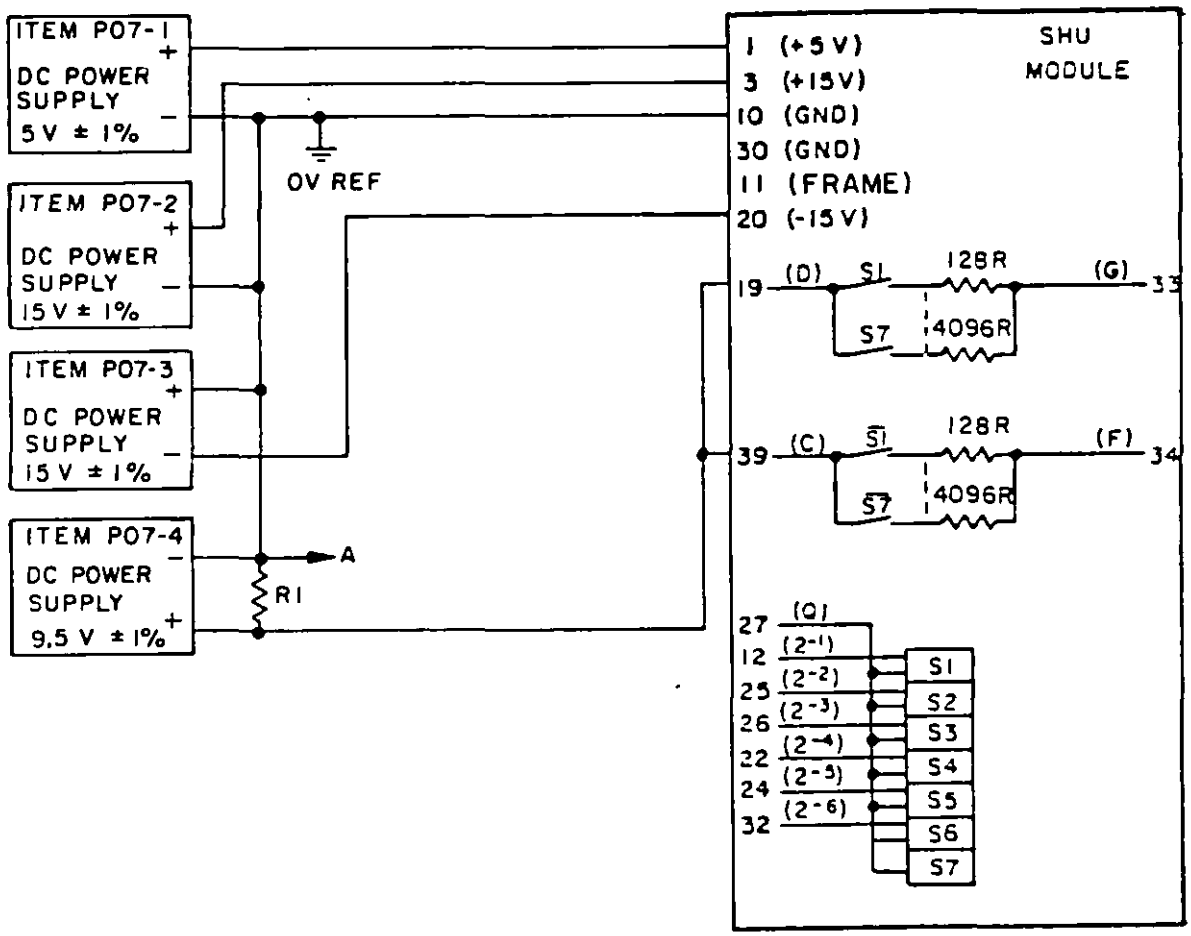
FIGURE 9. Hot socket check test circuit for table IV - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:32Z
Check the source to verify that this is the current version before use.

TABLE V. Switch check test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
D1	Switch check	Open	See figure 10, note 2			10
D2	"	27	"			"
D3	"	12,22,24,25,26	"			"
D4	"	12,22,25,26,32	"			"
D5	"	12,24,25,26,32	"			"
D6	"	12,22,24,25,32	"			"
D7	"	12,22,24,26,32	"			"
D8	"	22,24,25,26,32	"			"
D9	"	12,22,24,25,26,27	"			"
D10	"	12,22,25,26,27,32	"			"
D11	"	12,24,25,26,27,32	"			"
D12	"	12,22,24,25,27,32	"			"
D13	"	12,22,24,26,27,32	"			"
D14	"	22,24,25,26,27,32	"			"

Source: https://assist.dla.mil -- Downloaded: 2020-10-20T16:32Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 50Ω ±5%, 2 W.
2. This test is a design stress test. No measurements are made during test, but the module must meet all X type tests after subjection to test conditions D1-D14.
3. For test conditions D1-D14, point A shall be connected as specified in table V for a minimum of 5 seconds.

FIGURE 10. Switch check test circuit for table V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:32Z
Check the source to verify that this is the current version before use.

TABLE VI. V_S stress check test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
E1	V_S stress check	12,22,24,25,26,27, 32	359.30	0.70	Degrees	11
E2	V_S stress check	12,22,24,25,26,32	89.30	90.70	Degrees	11

Source: https://assist.dla.mil -- Downloaded: 2023-10-20T16:32Z
Check the source to verify that this is the current version before use.

TABLE VII. Angular accuracy (400 Hz) check test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
F1	Angular accuracy (400 Hz) check	12,22,24,25,26,27, 32	359.30	0.70	Degrees	11
F2	"	12,22,24,25,26,27	0.21	1.61	"	"
F3	"	12,22,25,26,27	2.12	3.52	"	"
F4	"	12,25,26,27	6.30	7.70	"	"
F5	"	12,25,27	16.32	17.72	"	"
F6	"	12,27	42.51	43.91	"	"
F7	"	27	88.39	89.79	"	"
F8	"	27,32	87.45	88.85	"	"
F9	"	24,27,32	85.49	86.89	"	"
F10	"	22,24,27,32	81.17	82.57	"	"
F11	"	22,24,26,27,32	70.87	72.27	"	"
F12	"	22,24,25,26,27,32	44.30	45.70	"	"
F13	"	12,22,24,25,26	88.39	89.79	"	"
F14	"	12,22,25,26	86.48	87.88	"	"
F15	"	12,25,26	82.30	83.70	"	"
F16	"	12,25	72.28	73.68	"	"
F17	"	12	46.09	47.49	"	"
F18	"	Open	0.21	1.61	"	"
F19	"	32	1.05	2.45	"	"
F20	"	24,32	3.01	4.41	"	"
F21	"	22,24,32	7.33	8.73	"	"
F22	"	22,24,26,32	17.63	19.03	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:32Z
Check the source to verify that this is the current version before use.

TABLE VII. Angular accuracy (400 Hz) check test - Continued.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
F23	Angular accuracy (400 Hz) check	22,24,25,26,32	44.30	45.70	Degrees	11
F24	Angular accuracy (400 Hz) check	12,22,24,25,26,32	89.30	90.70	Degrees	11

Source: https://assist.dla.mil -- Downloaded: 20261O-2O116:32Z
Check the source to verify that this is the current version before use.

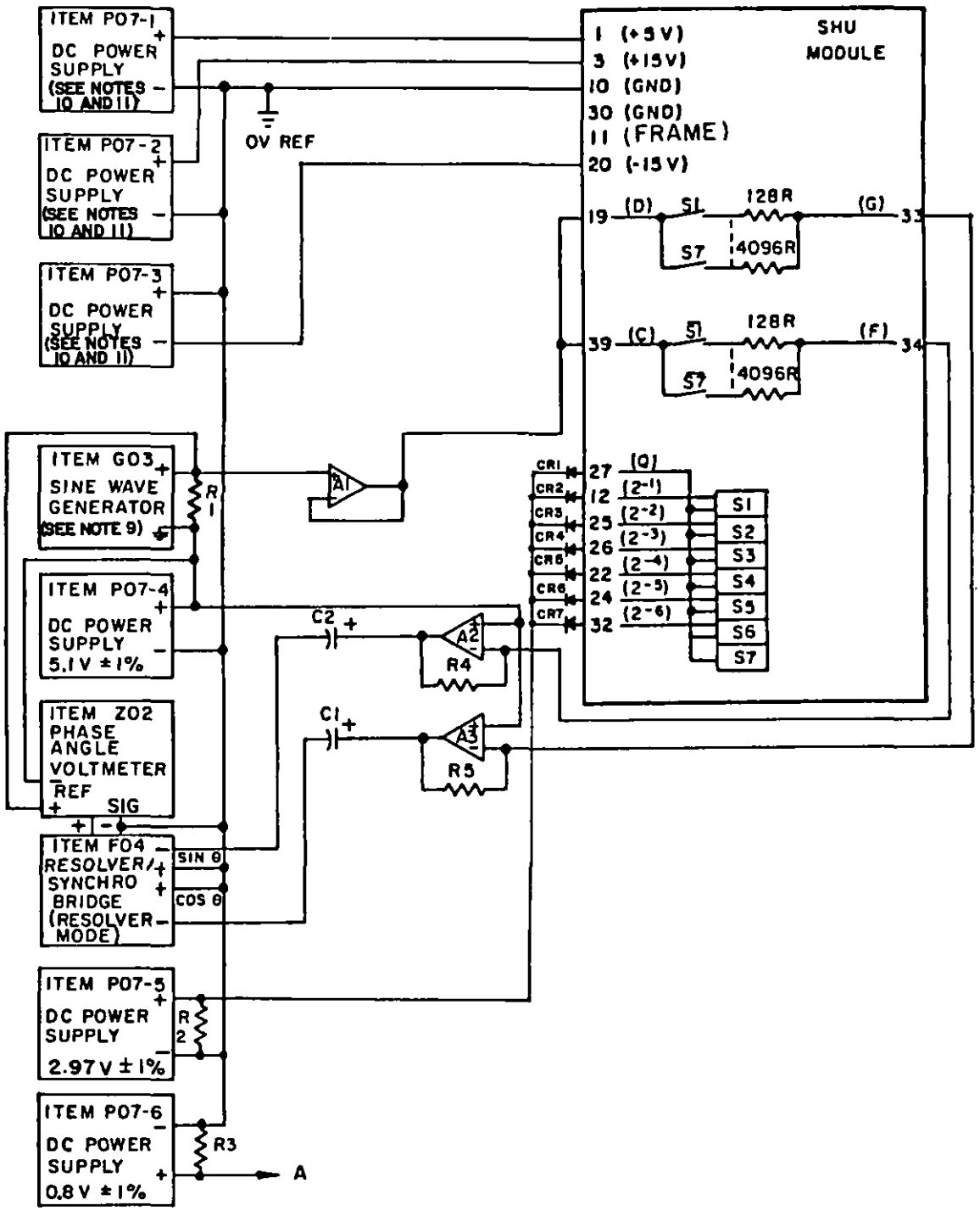


FIGURE 11. V_S stress check and angular accuracy (400 Hz) check test circuit for tables VI and VII.

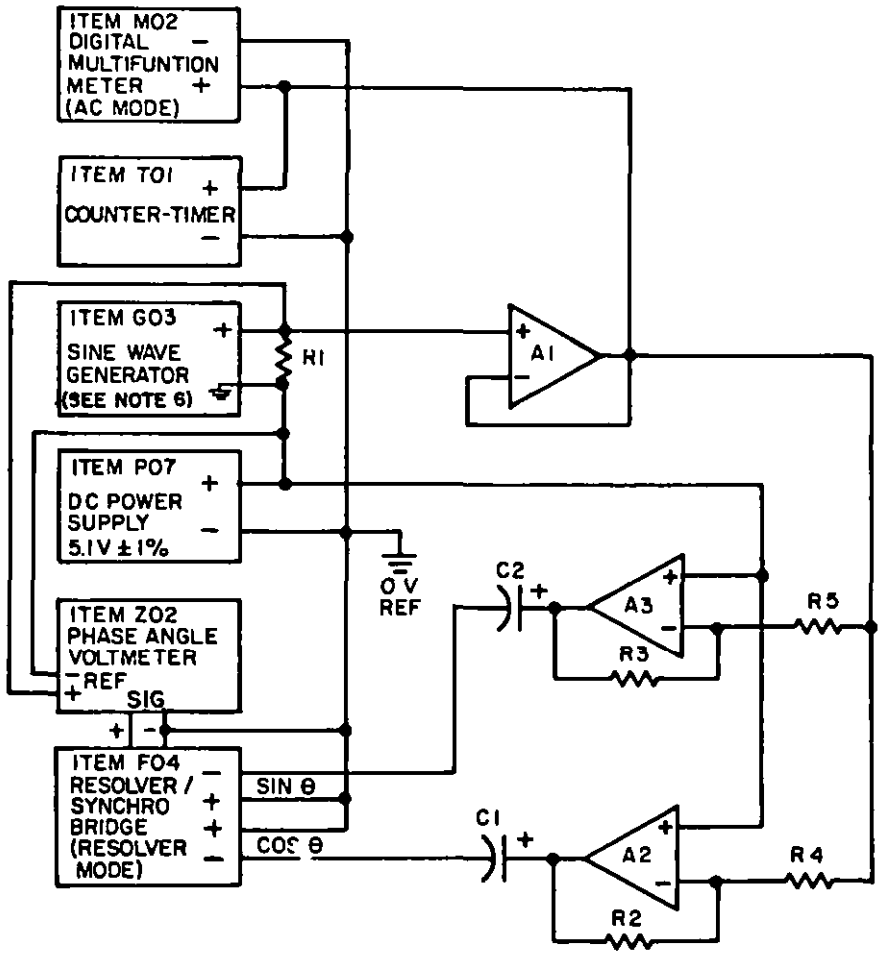
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:32Z
Check the source to verify that this is the current version before use.

NOTES:

1. R1 = 600Ω ±5%, 1 W.
2. R2 = 50Ω ±5%, 1 W.
3. R3 = 5Ω ±5%, 1/2 W.
4. R4 = R5 = 64 kΩ ±0.02% (matched to ±0.005%), 1/8 W.
5. C1 = C2 = 100 µF ±5%, 20 WV dc, electrolytic.
6. CR1-CR7 = 1N4150, or equivalent.
7. A1-A3 = µA741, or evquivalent.
8. Amplifiers A2 and A3 are to be matched per figure 12.
9. Item G03 setup (verified at pins 19 and 39):
   Amplitude = 3.18 ±0.01 V rms.
   Frequency = 400 ±4 Hz.
10. The V_S stress check test verifies correct module operation after the module power inputs were stressed for a minimum of 5 seconds and a maximum of 30 seconds.
    (a) Initially adjust item P07-1 for 7 V ±1% and items P07-2 and P07-3 for 18 V ±1%.
    (b) Without disconnecting the test circuit, decrease the voltage of item P07-1 to 5 V ±1% and items P07-2 and P07-3 to 15 V ±1%.
    (c) For test condition E1, null item F04 and record reading.
    (d) Without power interruption, make E2 measurement.
11. Test condition F verifies an angular accuracy of ±0.70 degrees.
    (a) Adjust power item P07-1 for 4.5 V ±1% and items P07-2 and P07-3 for 13.5 V ±1%.
    (b) Null item F04 and record reading.

FIGURE 11. V_S stress check and angular accuracy (400 Hz) check test circuit for tables VI and VII - Continued.

Source: https://assist.dla.mil -- Downloaded: 202610-20T16:32Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 600Ω ±5%, 1 W.
2. R2 = R3 = 64 kΩ ±0.02% (matched to ±0.005%), 1/8 W.
3. R4 = R5 = 64 kΩ ±1% (matched to ±0.005%), 1/8 W.
4. C1 = C2 = 100 µF ±5%, 20 WV dc, electrolytic.
5. A1-A3 = µA741, or equivalent.
6. Item GO3 setup (verified at output of A1):
   Amplitude = 3.20 ±0.01 V rms.
   Frequency = 400 ±4 Hz.
7. The null for item FO4 shall be between 44.99 and 45.01 degrees.
8. Set item PO7 for A1, A2, and A3 ±15 V ±1%.

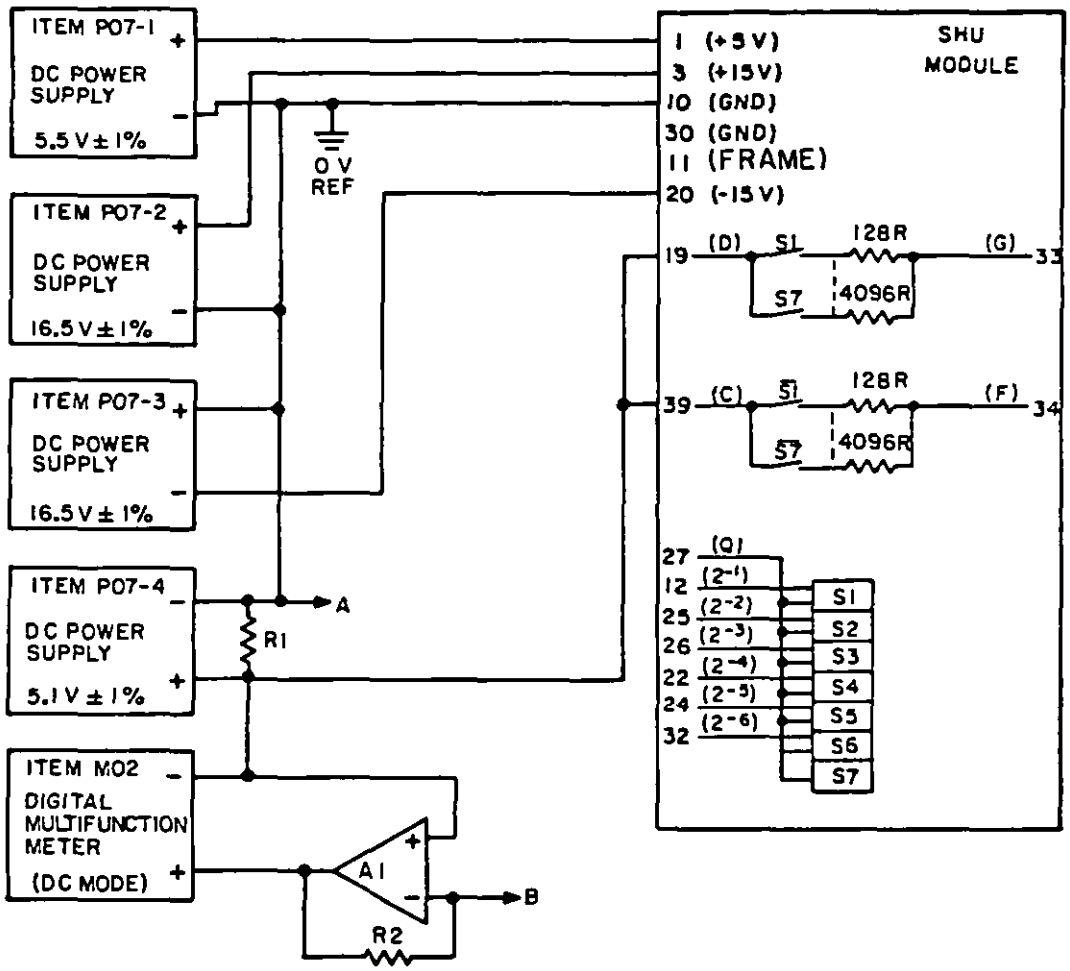
FIGURE 12. Test circuit to match A2 and A3.

Source: https://assist.dla.mil -- Downloaded: 20260-20T16:32Z
Check the source to verify that this is the current version before use.

TABLE VIII. DC integrity and leakage check test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
G1	DC integrity and leakage check	22,24,25,26,27,32	34	-0.4	1.6	V	13
G2	"	12,22,24,26,27,32	"	"	"	"	"
G3	"	12,22,24,25,27,32	"	"	"	"	"
G4	"	12,24,25,26,27,32	"	"	"	"	"
G5	"	12,22,25,26,27,32	"	"	"	"	"
G6	"	12,22,24,25,26,27	"	"	"	"	"
G7	"	Open	"	"	"	"	"
G8	"	12,22,24,25,26,32	"	"	"	"	"
G9	"	12,27	33	"	"	"	"
G10	"	25,27	"	"	"	"	"
G11	"	26,27	"	"	"	"	"
G12	"	22,27	"	"	"	"	"
G13	"	24,27	"	"	"	"	"
G14	"	32,27	"	"	"	"	"
G15	"	27	"	"	"	"	"
G16	"	12,22,24,25,26,27,32	"	"	"	"	"
G17	"	12,22,24,25,26,32	"	"	0.8	"	"
G18	"	12,22,24,25,26,27,32	34	"	0.8	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:32Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. R2 = 4 MΩ ±1%, 1/8 W.
3. A1 = AD233K, or equivalent.
4. DC offset of A1 shall be less than 0.1 mV.
5. This test verifies that each switch has less than 1.5 mV of DC offset.
6. Record measurement as read on item M02.

FIGURE 13. DC integrity and leakage check test circuit for table VIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:32Z
Check the source to verify that this is the current version before use.

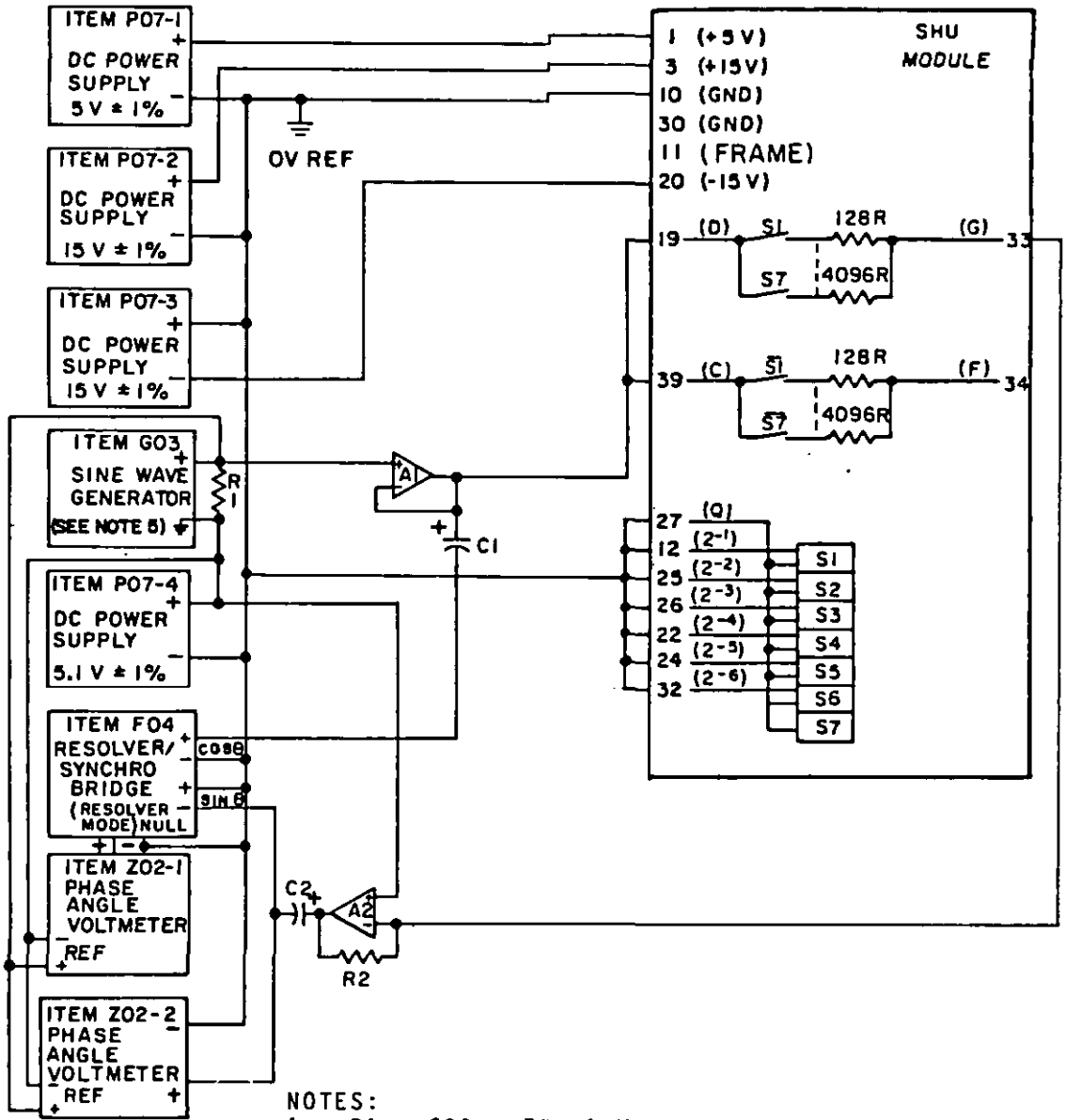
TABLE IX. Absolute transformation ratio check and phase shift test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
H1	Absolute transformation ratio check	44.8564	45.1429	Degrees	14
H2	Phase shift	179.5	180.5	Degrees	14

TABLE X. Input current (analog) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J1	Input current (analog), V_I = 4.6 V	19	39	10	0	50	µA	15
J2	"	39	19	10	"	"	"	"
J3	"	39	19	1	"	"	"	"
J4	"	19	39	1	"	"	"	"

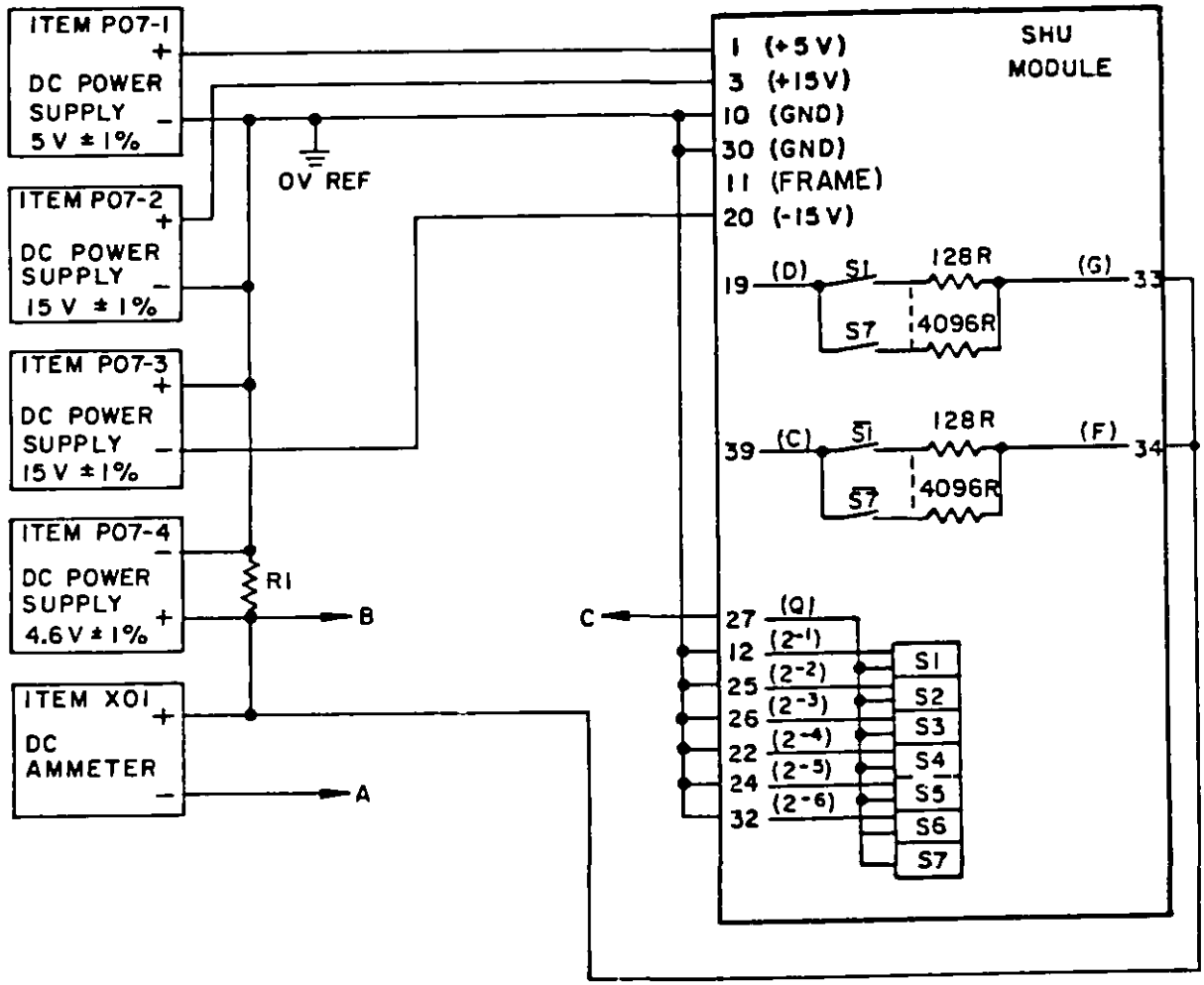
Source: https://assist.dla.mil -- Downloaded: 2026-01-20T16:32Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 600Ω ±5%, 1 W.
2. R2 = 64 kΩ ±0.02%, 1/8 W.
3. C1 = C2 = 100 µF ±5%, 20 WV dc, electrolytic.
4. A1 = A2 = µA741, or equivalent.
5. Item G03 setup:
   Amplitude -= 3.18 ±0.01 V rms.
   Frequency = 400 ±4 Hz.
6. For test condition H1, read item F04.
7. For test condition H2, read item Z02-2.

FIGURE 14. Absolute transformation ratio check and phase shift test circuit for table IX.

Source: https://assist.dla.mil -- Downloaded: 20260-20T16:32Z
Check the source to verify that this is the current version before use.



CAUTION
Do not isolate the chassis of item X01 from earth ground.

NOTE: R1 = 50Ω ±5%, 1 W.

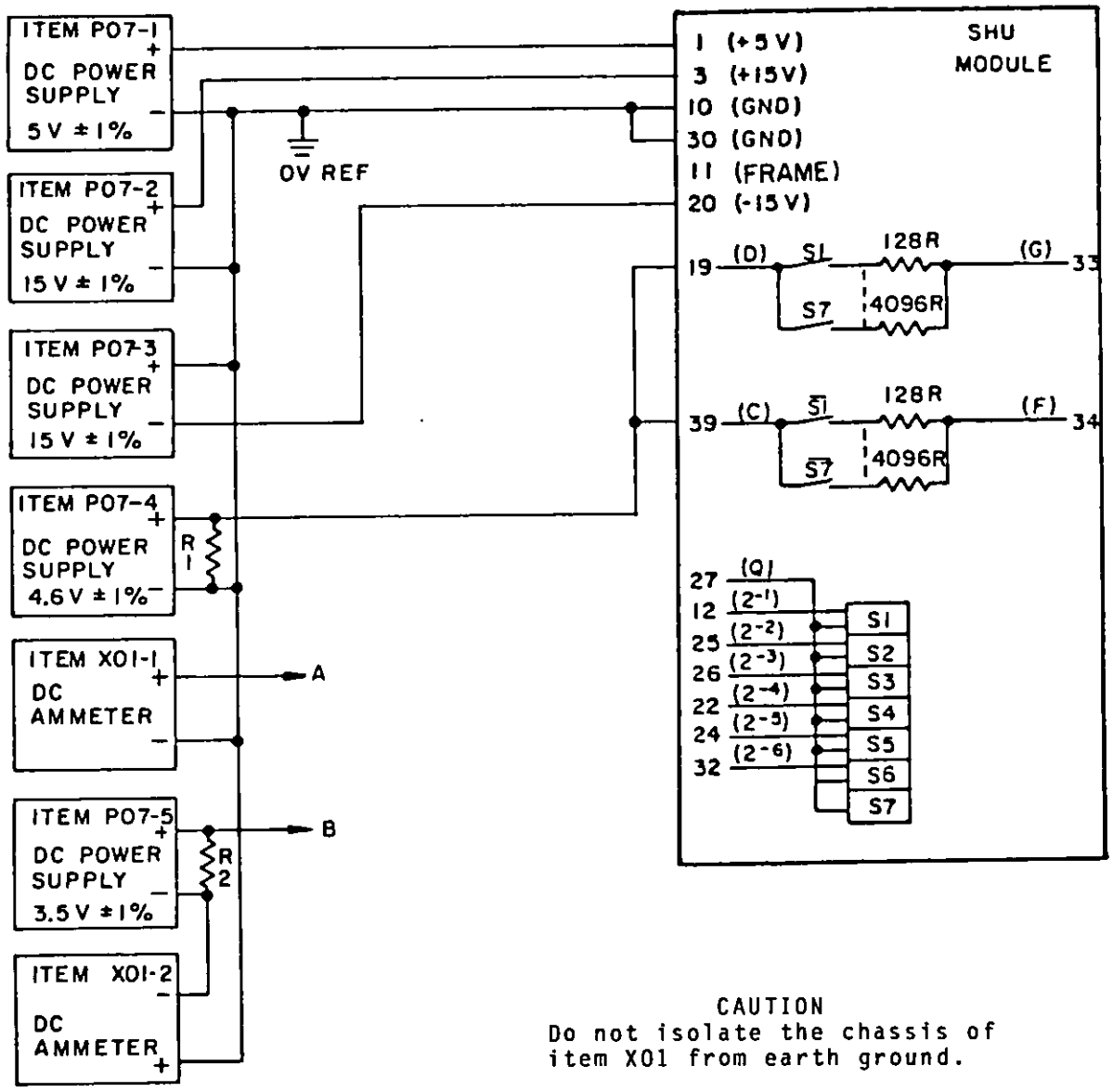
FIGURE 15. Input current (analog) test circuit for table X.

Source: https://assist.dla.mil Downloaded: 2026-10-20T16:32Z
Check the source to verify that this is the current version before use.

TABLE XI. Input current (low-level and high level) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K1	2^{-1} Input current, low level	12	Open	0.9	1.4	mA	16
K2	2^{-2} "	25	"	"	"	"	"
K3	2^{-3} "	26	"	"	"	"	"
K4	2^{-4} "	22	"	"	"	"	"
K5	2^{-5} "	24	"	"	"	"	"
K6	2^{-6} "	32	"	"	"	"	"
K7	Q "	27	"	225	275	µA	"
K8	2^{-1} Input current, high level	Open	12	275	440	"	"
K9	2^{-2} "	"	25	"	"	"	"
K10	2^{-3} "	"	26	"	"	"	"
K11	2^{-4} "	"	22	"	"	"	"
K12	2^{-5} "	"	24	"	"	"	"
K13	2^{-6} "	"	32	"	"	"	"
K14	Q "	"	27	68	84	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:32Z
Check the source to verify that this is the current version before use.



CAUTION
Do not isolate the chassis of item XO1 from earth ground.

NOTES:
1. R1 = R2 = 50Ω ±5%, 1 W.
2. Since items XO1-1 and XO1-2 are not used simultaneously, only one item XO1 is required.
3. With point "B" disconnected, item XO1-2 should indicate zero current on the 1 μA range.

* FIGURE 16. Input current (low-level and high-level) test circuit for table XI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:32Z
Check the source to verify that this is the current version before use.

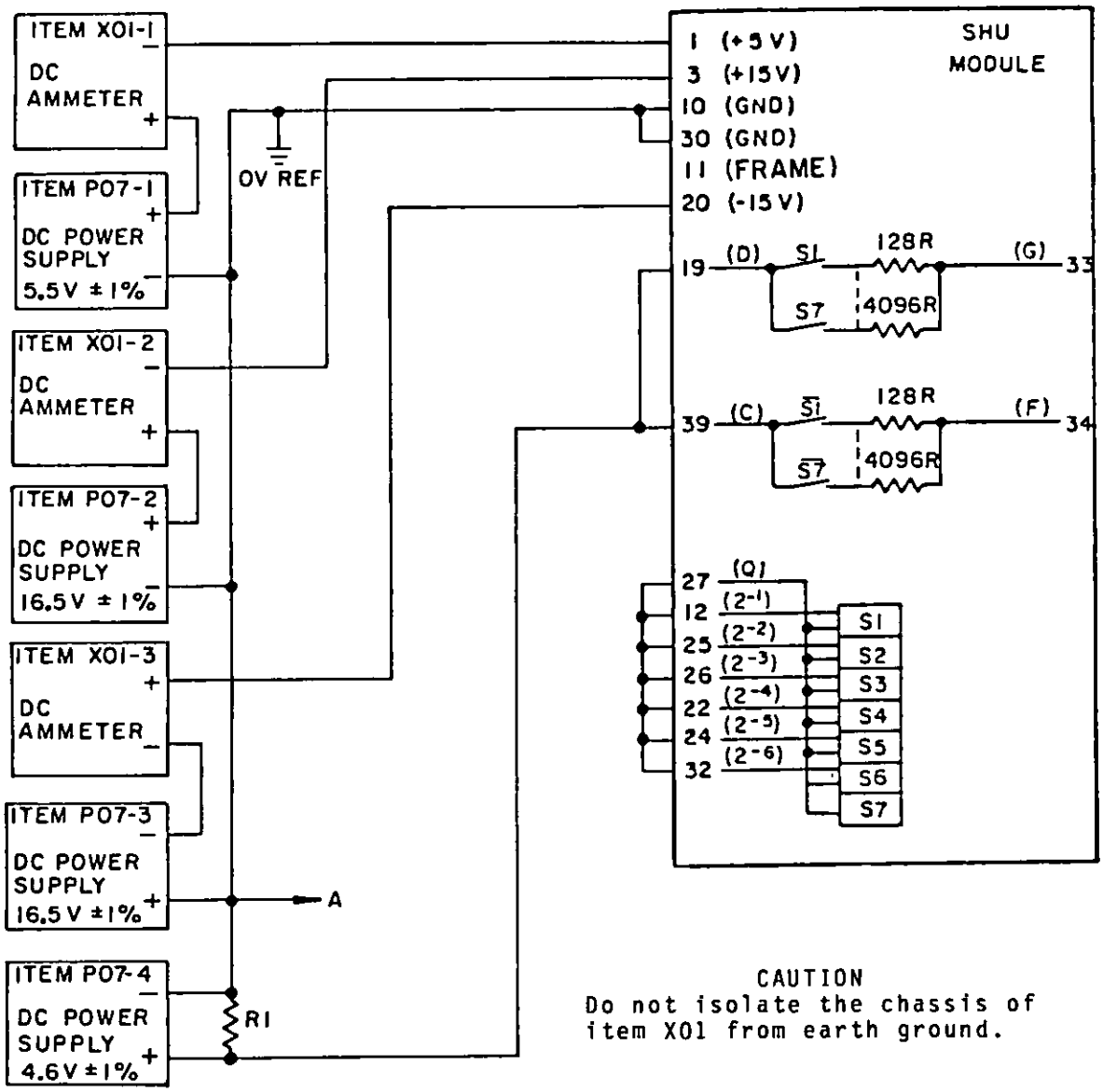
TABLE XII. Power supply current test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
L1	Power supply current, 5.5 V	Open	0	3	mA	17
L2	Power supply current, + 16.5 V	"	"	5	"	"
L3	Power supply current, - 16.5 V	"	"	5	"	"
L4	Power supply current, 5.5 V	27	6.5	15.0	"	"

TABLE XIII. Continuity test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M1	Continuity, ground	10	30	0.0	0.1	Ω	18
M2	Continuity, ground	10	30	0	1	Ω	18
M3	Continuity, frame	11	Frame	0	1	Ω	18

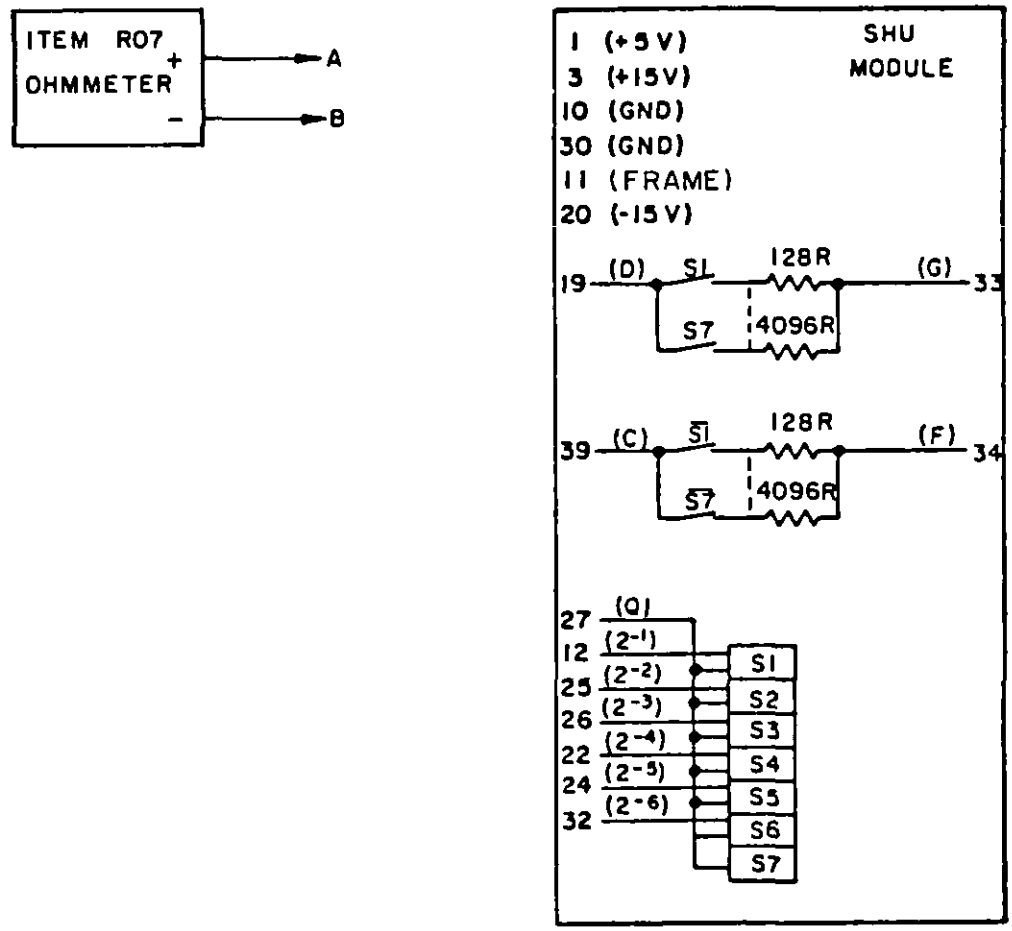
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:32Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. Since items XO1-1, XO1-2, and XO1-3 are not used simultaneously, only one item XO1 is required.

FIGURE 17. Power supply current test circuit for table XII.

Source: https://assist.dla.mil Downloaded: 2026-10-20T16:32Z
Check the source to verify that this is the current version before use.



NOTE:
For test condition M3, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

* FIGURE 18. Continuity test circuit for table XIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:32Z
Check the source to verify that this is the current version before use.

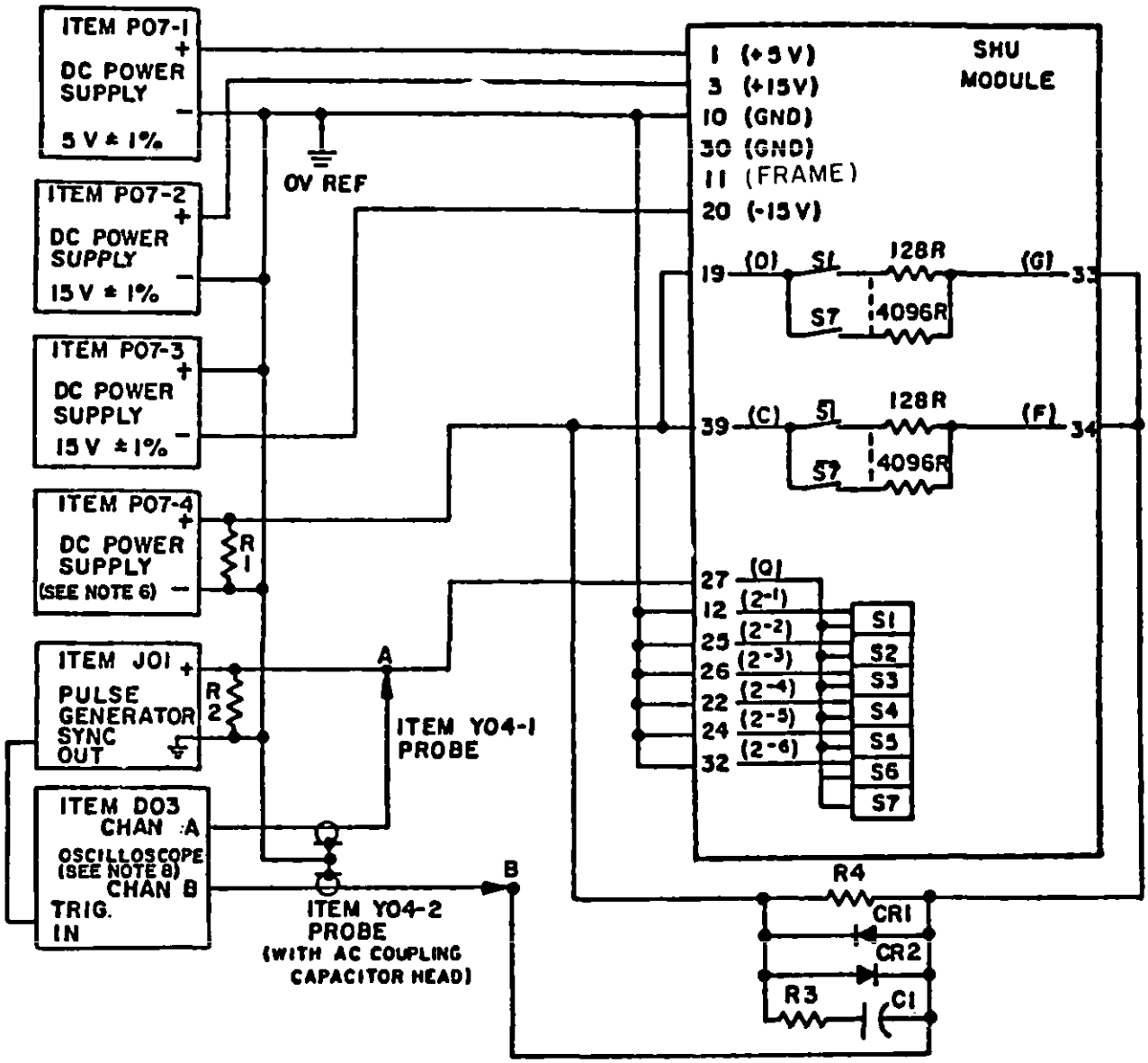
TABLE XIV. Settling time test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
N1	Settling time (t_{S1})	0	4	µs	19,20
N2	Settling time (t_{S2})	"	"	"	"
N3	Settling time (t_{S1})	"	"	"	"
N4	Settling time (t_{S2})	"	"	"	"

TABLE XV. Angular accuracy (800 Hz) test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
P1	Angular accuracy, 800 Hz	22, 25, 32	26.950	28.350	Degrees	21
P2	"	22, 25, 27, 32	61.650	63.050	"	"
P3	"	12, 24, 26, 27	25.330	26.730	"	"

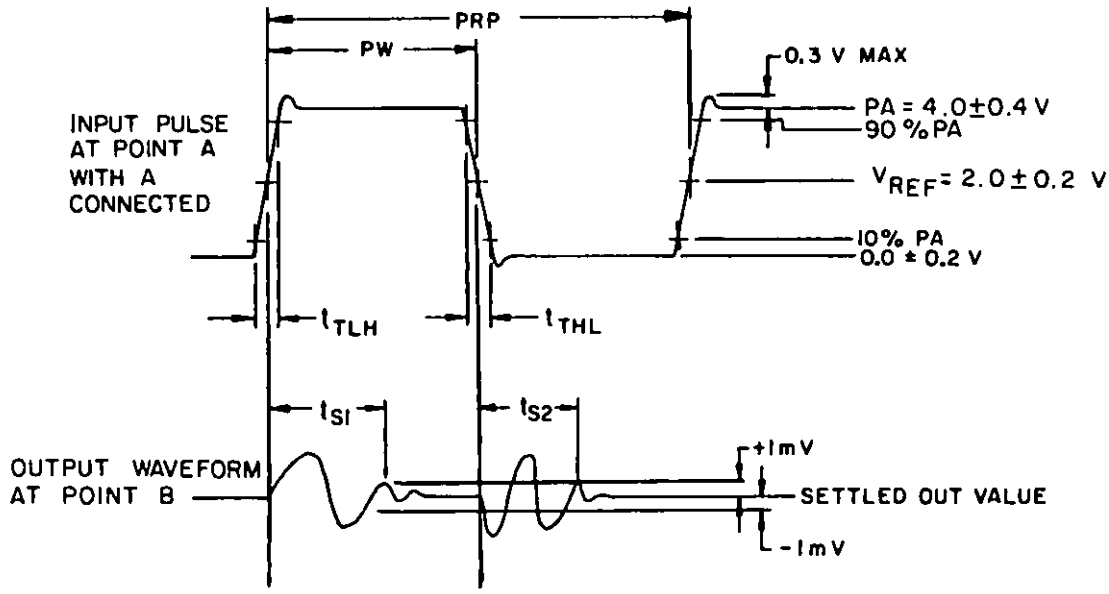
Source: https://assist.dla.mil -- Downloaded: 2020-10-20T16:32Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = R2 = 50Ω ±5%, 1 W.
2. R3 = 750Ω ±1%, 1/8 W.
3. R4 = 777Ω ±0.1%, 1/8 W.
4. C1 = 220 pF ±5%, 20 WV dc.
5. CR1 = CR2 = 1N4148, or equivalent.
6. For test conditions N1 and N2, adjust item P07-4 for 6 V ±1% and for test conditions N3 and N4, adjust item P07-4 for 1.1 V ±1%.
7. Item D03 shall be ac coupled so that channel B can be amplified for maximum amplitude or a dc voltage can be subtracted by means of a differential comparator. Verify that channel B output amplitude stays within ±1% mV of its settled out value at 4 µs after the input pulse.
8. Use type W plug-in.

FIGURE 19. Settling time test circuit for table XIV.

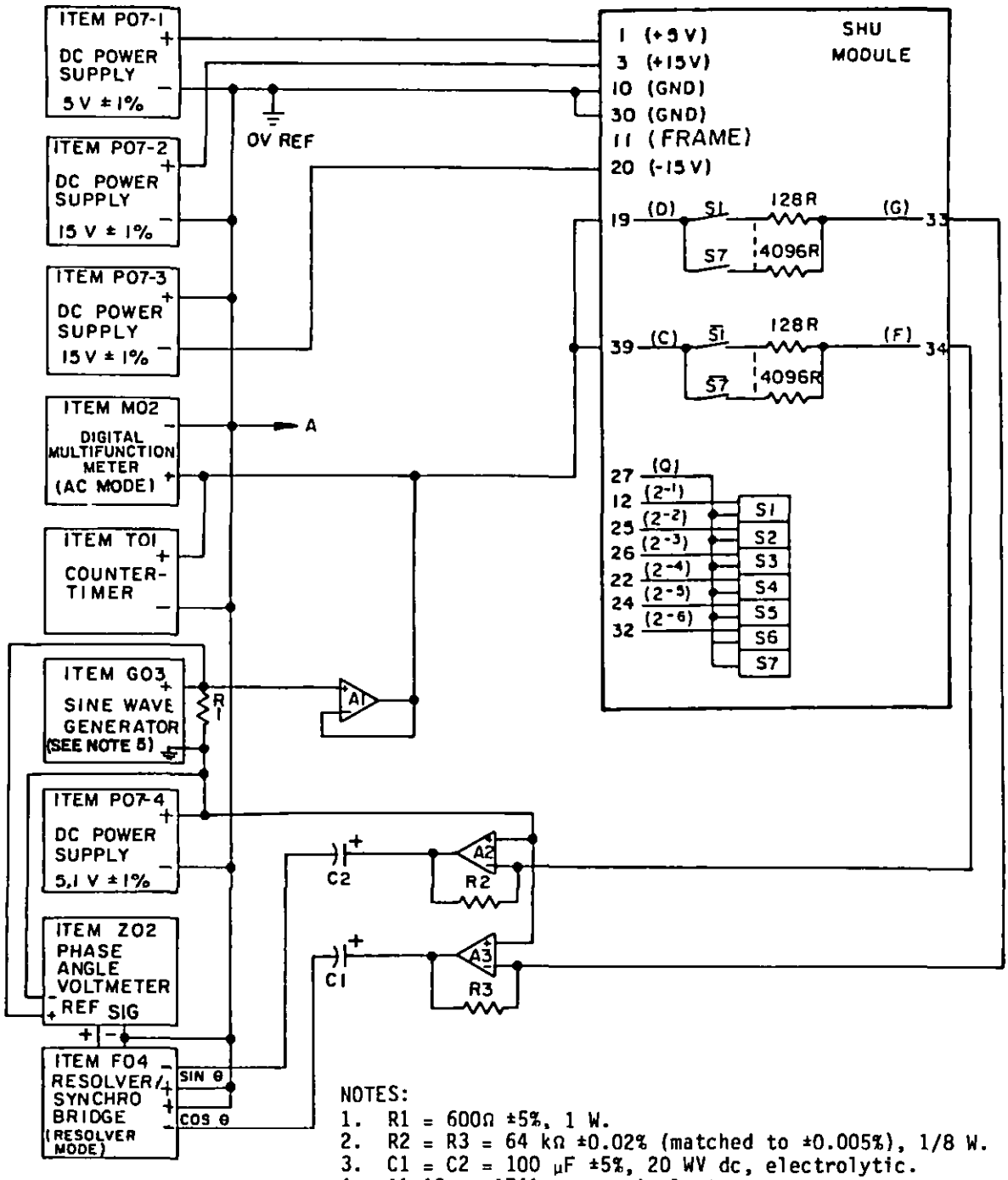
Source: https://assist.dla.mil -- Downloaded: 202610-20T16:32Z
Check the source to verify that this is the current version before use.



NOTES:
1. Input pulse parameters:
   PW = 10.0 ±0.5 μs.
   PRP = 20 ±1 μs.
   $t_{TLH} = t_{THL} = 50 \pm 5$ ns.
2. $t_{S1} = t_{S2} \leq 4.0$ μs.

FIGURE 20. Settling time test circuit waveforms for table XIV.

Source: https://assist.dla.mil -- Downloaded: 20261O-20T16:32Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 600Ω ±5%, 1 W.
2. R2 = R3 = 64 kΩ ±0.02% (matched to ±0.005%), 1/8 W.
3. C1 = C2 = 100 µF ±5%, 20 WV dc, electrolytic.
4. A1-A3 = µA741, or equivalent.
5. Item G03 setup (verified at pins 19 and 39):
   Amplitude = 3.2 ±0.1 V rms.
   Frequency = 800 ±8 Hz.

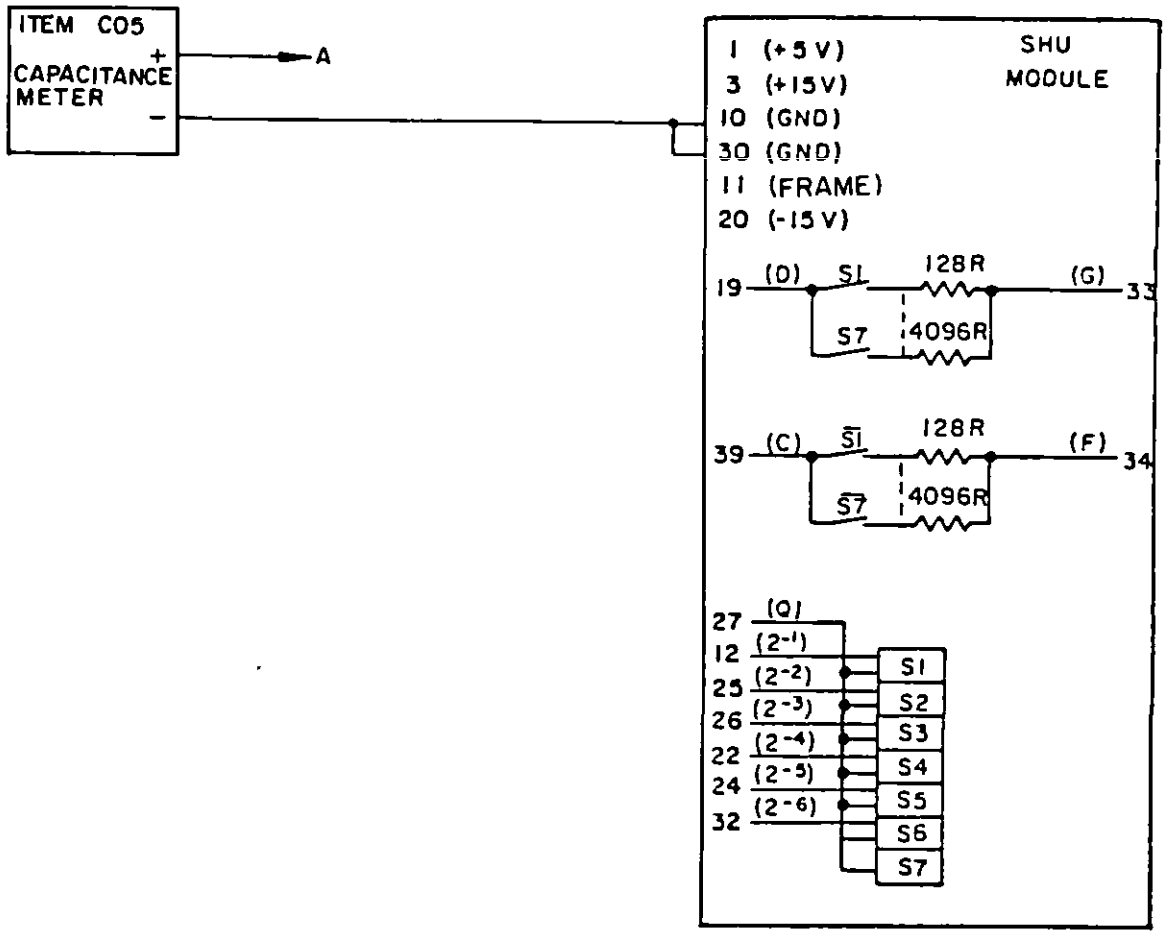
FIGURE 21. Angular accuracy (800 Hz) test circuit for table XV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:32Z
Check the source to verify that this is the current version before use.

TABLE XV. Input and output capacitance test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
Q1	2^{-1} Input capacitance	12	2	35	pF	22
Q2	2^{-2} "	25	"	"	"	"
Q3	2^{-3} "	26	"	"	"	"
Q4	2^{-4} "	22	"	"	"	"
Q5	2^{-5} "	24	"	"	"	"
Q6	2^{-6} "	32	"	"	"	"
Q7	Q "	27	"	"	"	"
Q8	G Output capacitance	33	50	200	"	"
Q9	F Output capacitance	34	50	200	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:32Z
Check the source to verify that this is the current version before use.



NOTES:
1. Null out effect of the test leads prior to connecting to the module.
2. Do not attach test connector or wires to any of the module pins other than pins 10 and 30 and the pin under test. Stray capacitance added to other pins may affect the readings.

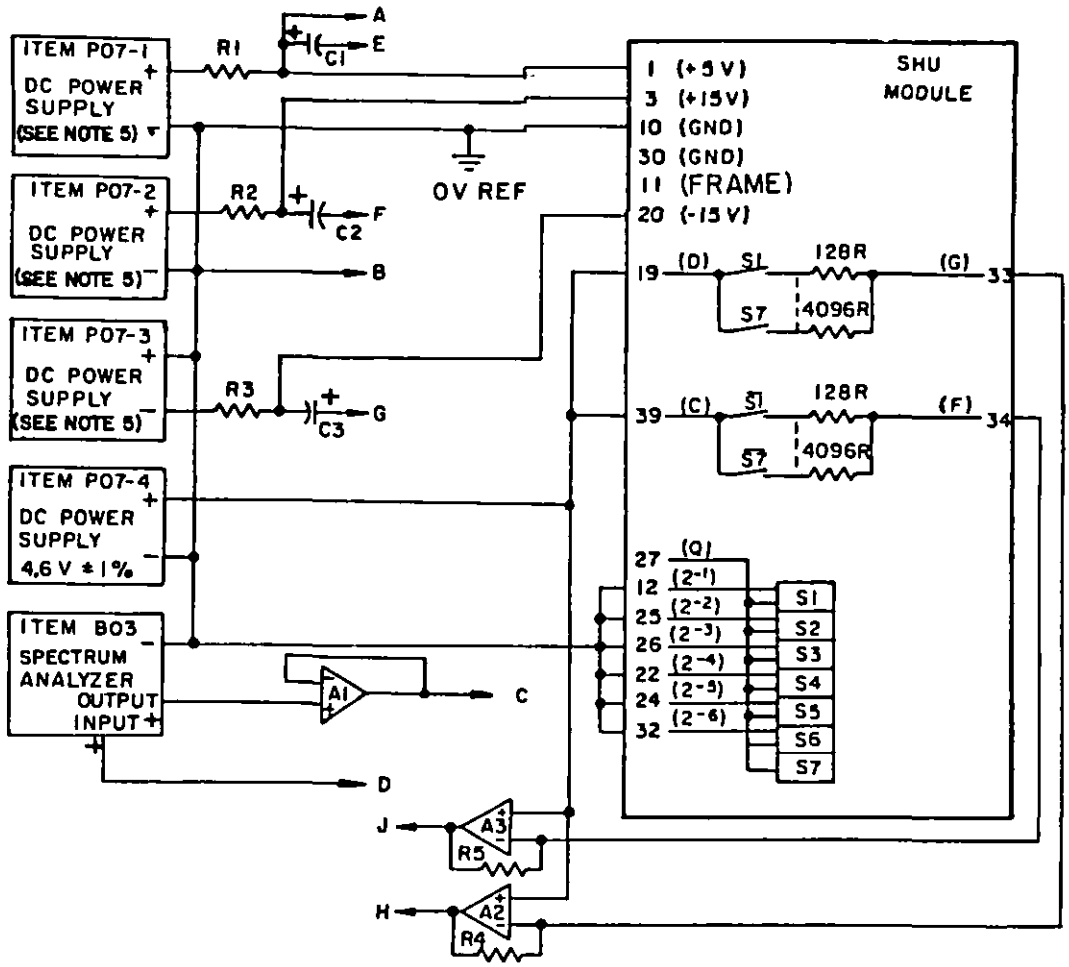
FIGURE 22. Input and output capacitance test circuit for table XVI.

Source: https://assist.dla.mil -- Downloaded: 202640-20T16:32Z
Check the source to verify that this is the current version before use.

TABLE XVII. Power supply noise rejection test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R1	Power supply noise rejection, 5 V	27	Open	E	H	0	1	mV rms	23
R2	Power supply noise rejection, 5 V	Open	27	E	J	"	"	"	"
R3	Power supply noise rejection, + 15 V	27	Open	F	H	"	"	"	"
R4	Power supply noise rejection, + 15 V	Open	27	F	J	"	"	"	"
R5	Power supply noise rejection, - 15 V	27	Open	G	H	"	"	"	"
R6	Power supply noise rejection, - 15 V	Open	27	G	J	"	"	"	"
R7	Power supply noise rejection, 5 V	27	Open	E	H	"	6	"	"
R8	Power supply noise rejection, 5 V	Open	27	E	J	"	"	"	"
R9	Power supply noise rejection, + 15 V	27	Open	F	H	"	"	"	"
R10	Power supply noise rejection, + 15 V	Open	27	F	J	"	"	"	"
R11	Power supply noise rejection, - 15 V	27	Open	G	H	"	"	"	"
R12	Power supply noise rejection, - 15 V	Open	27	G	J	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T16:32Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1-R3 = 100Ω ±5%, 1/8 W.
2. R4 = R5 = 64 kΩ ±1% (matched to ±0.002%), 1/8 W.
3. C1-C3 = 100 µF ±5%, 20 WV dc, electrolytic.
4. A1-A4 = µA741, or equivalent.
5. Power supply setups:
   P07-1 = 5 V ±1% (verified at pin 1).
   P07-2 = 15 V ±1% (verified at pin 3).
   P07-3 = -15 V ±1% (verified at pin 20).
6. Item B03 output:

Amplitude	Frequency	Test condition
0.354 V rms ±5%	50 Hz to 1 kHz	R1 and R2
0.707 V rms ±5%	50 Hz to 1 kHz	R3 and R4
0.707 V rms ±5%	50 Hz to 1 kHz	R5 and R6
0.354 V rms ±5%	1 kHz to 10 kHz	R7 and R8
0.707 V rms ±5%	1 kHz to 10 kHz	R9 thru R12

7. Item B03 bandwidth is to be 10 Hz.

FIGURE 23. Power supply noise rejection test circuit for table XVII.

Source: https://assist.dla.mil -- Downloaded: 20260-20T16:32Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced *document(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY

Naval Electronic Systems Command
Washington, DC 20363

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, DC 20363
ATTN: ELEX 8111

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:32Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-28787/187B(EC)	SEM Generator, Function, LSB, Analog, Key Code SHU

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

Source: https://assist.dla.mil -- Downloaded: 20260120T16:32Z
Check the source to verify that this is the current version before use.